App Note – HVLP & Dragon8 & Gen5 Encoder Configuration

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DOCUMENT APPROVAL

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1 Introduction

1.1 Purpose and Scope

The scope of this document is to explain the configuration for encoder on HVLP, Dragon8 and Gen5.

1.2 Related documents

Ref	Doc Ref	Title	Origin	Revision
1				
2				

Table 1. Related Documents

1.3 Document Conventions

Typeface	Indicates
WARNING	Very important information for the document user. Use this to indicate instructions or constraints the document user must comply with to prevent serious problems.
CAUTION	Important information for the document user. Use this to indicate instructions or constraints the document user must comply with to prevent undesirable problems.
NOTE	Useful information for the document user.

Table 2. Typographical Conventions

1.4 Glossary

Term	Description

Table 3. Glossary

1.5 Conditions for document usage

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2 Introduction

This document describes the encoder system used by 'Genus' firmware. 'Genus' is a term used to describe the software present in HVLP, Dragon8 and Gen5 size9. These products share some generic code and functionality, for example TracApp and NVM - as well as some specific modules only applicable to themselves.

Genus products differ from conventional Gen4 in that they feature a new encoder system, based around a second microcontroller, namely the IO processor (IOP). The dedicated IO processor allows for far higher encoder control bandwidth than Gen4 and also allows software implemented resolver control method rather than a dedicated integrated circuit. Control and configuration remains the same as Gen4 in that all parameters are entered to the Motor Control Processor (MCP) and the IOP is only accessed directly via CAN in the bootloader when uploading new firmware (it is typically allocated node 17 but this is configurable in 0x5900, sub 3).

As resolver modulation/demodulation is now handled by the IOP, genus now supports a software-adjustable gain for the resolver, so each product can support a wide range of gains without hardware modification (e.g. 0.125x, 0.25x. 0.5x). Unlike Gen4 size 8/10 where different hardware part numbers are required for each resolver type.

User configuration remains very similar to Gen4, as settings are still input via the object dictionary; which contains descriptions and scaling of all the new settings. However, the encoder settings now reside in new objects (see Figure 1) and the functionality and performance have increased.

The encoder type is specified in object 0x4631 as well as filtering and other common features to all encoders. The items specific to each encoder type are contained in objects 0x4632 to 0x4635 as described in Figure 1.

Four encoder types are supported:

- Resolver
- Sine-cosine
- UVW
- AB

Any encoder type can be allocated to any motor type without constraints, so it is up to the user to determine the suitability (e.g. PMAC requires an absolute position sensor, AB is incremental).

Suitable combinations:

PMAC Geometric or Lookup-table

- Resolver
- Sine-Cosine
- UVW

Induction Flux-Vector

- AB
- Resolver
- Sine-Cosine

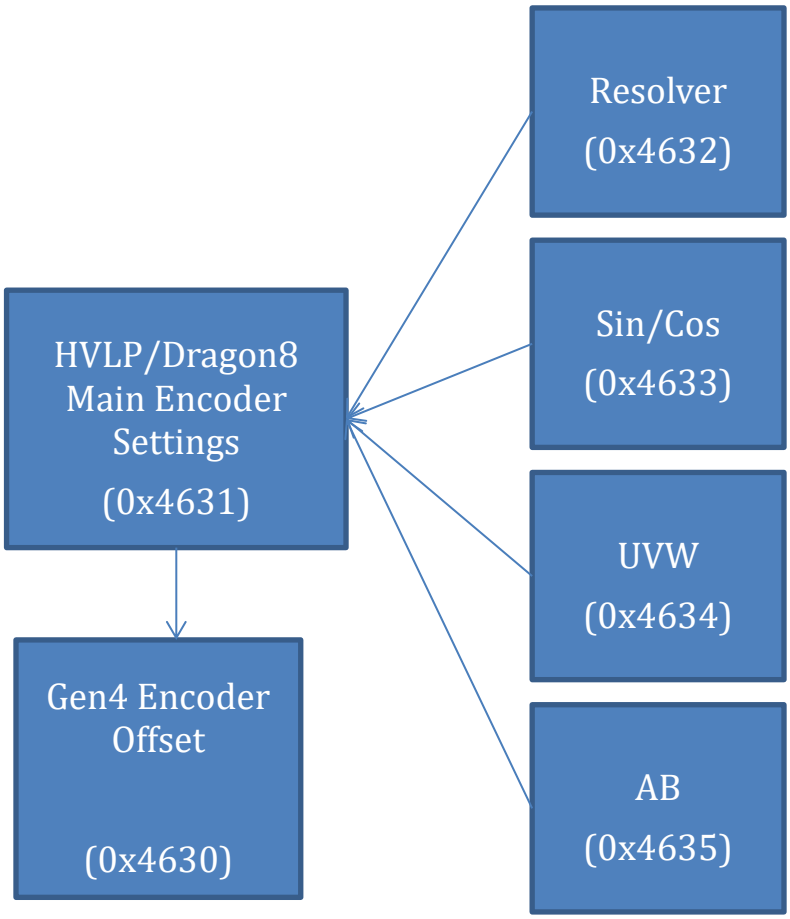


Figure 1 – Object dictionary parameter layout of encoders and encoder specific parameters

3 Encoder Settings (0x4630)

Most Gen4 items no longer exist in object 0x4630 as they have been relocated to object 0x4631 to 0x4635 to clarify configuration required based on encoder type. However, the remaining 0x4630 items that can be modified are (see Figure 2):

- sub 2: the user can select between 5V and 10V supply for products which support this (HVLP, Gen5 size 9).
- sub 4: encoder offset is also present. Encoder offset is scaled to 0.0549332 degrees per bit, so -180 to +180 can be selected. This allows all possible encoder offsets with the greatest precision in a 16bit parameter.

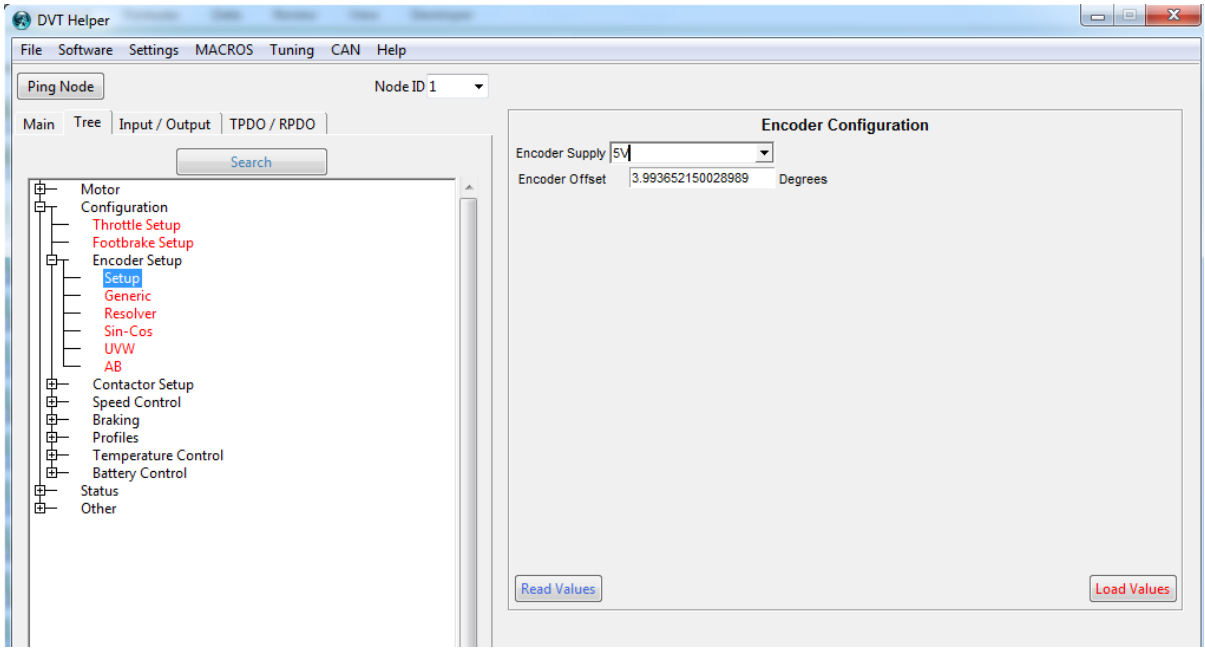


Figure 2 – 0x4630 Encoder Settings

NOTE	Dragon8 includes permanent 5V and 10V outputs on different pins which the user hard-wires to the encoder as required.

4 Genus Encoder Settings (0x4631)

4.1 Parameters

4.1.1 Encoder Type (0x4631, sub 1)

All of the Genus products support all encoder types and resolver gains. The encoder type can be configured in object 0x4631 as either:

- **0x00** Resolver absolute position sensor
- **0x01** Sine-Cosine absolute position sensor
- **0x02** UVW absolute position sensor
- **0x03** AB incremental position sensor

Combined UVW+AB is not supported in Genus, sin-cos or resolver are recommended instead

4.1.2 PLL gains (0x4631, sub 4-6)

Filtering is applied to all encoder types to reduce distortion of the angle and speed measurement used internally by the firmware. Please use the latest default object dictionary gains for these items.

- sub4: Gain 40000
- sub5: K1 65000
- sub6: K2 40000

Typically there should be no need to change the filter gains to suit an application. However, if an encoder appears noisy then consult BorgWarner for advice. Setting all 3 PLL gains to zero can be done to disable the PLL filter on the angle.

4.1.3 Encoder latency (0x4631, sub 10)

Latency compensation can be applied to any encoder type, although latency typically only affects Sin-Cos and UVW (to a lesser extent). Latency is the delay between the controller and the sensor in which it is expressed in microseconds. Please consult BorgWarner for suggested latencies to suit known sensor types.

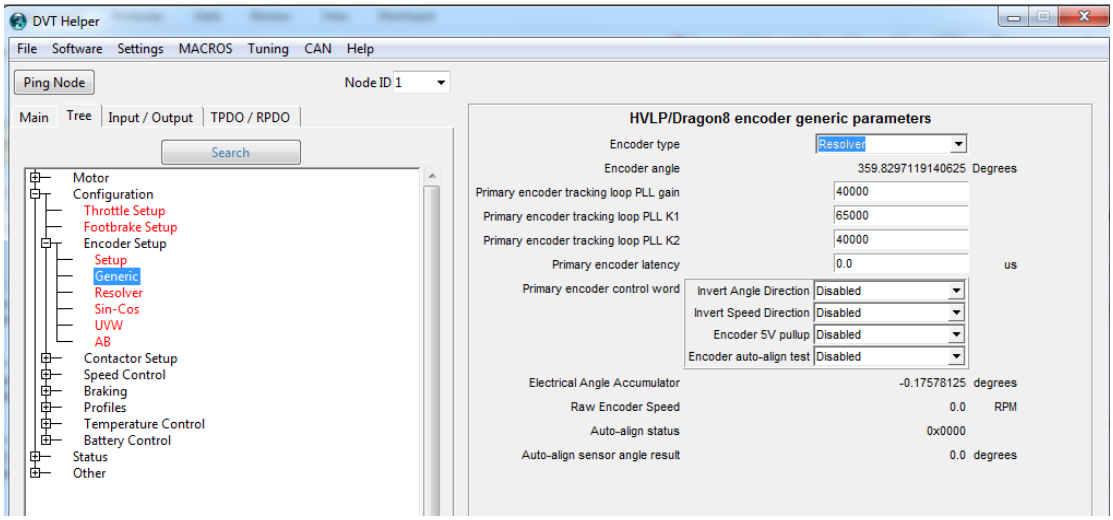


Figure 3 - Genus generic encoder parameters in DVT Helper

WARNING	Setting the latency compensation too high can cause undesireable motor operation. Set the value to 0µs until the impact of sensor latency is known

4.1.4 Encoder control word (0x4631, sub 11)

- 0x0001 (bit0) Invert angle direction. When an encoder is wired to rotate in the opposite direction to the motor, no rotation can occur and the rotor will lock (PMAC) or rotate at slip frequency (flux-vector). Rather than re-wiring the motor or encoder, the angle direction can be inverted here. Note that negative speed for positive torque requires a 180 degree change to the encoder offset, NOT the angle direction.
- *(0x0002 (bit1) If the speed direction invert option is listed then ignore as this is no longer used)*
- 0x0004 (bit2) Encoder pullup. On dragon8 this enables the pull-up circuitry for AB encoder type.
- 0x0008 (bit3) Auto align enable. For explanation on how to auto-align the encoder, see section 9.

4.1.5 Other encoder parameters

The user can also monitor the following debug items which are read-only:

- Electrical angle accumulator (0x4631, sub 13) – an accumulation of the electrical angle since power-on, up to ±32767 cycles (approx. $\pm 11796120^0$)
- The results of an encoder auto-align test (0x4631, sub 15-16). See section 9 for more information.

5 Resolver (0x4632)

5.1 Parameters

The Resolver settings reside in object 0x4632 and can also be modified in **Configuration -> Encoder Setup -> Resolver** in DVT helper. Check that the encoder type is set as 'Resolver' in the encoder type in 0x4631, 1 (0x00 = Resolver). See Figure 4 for the resolver parameters that can be modified/viewed in object 0x4632.

5.1.1 Status-Word (0x4632, sub 1)

The status word is read only but will indicate either of the following states if there are issues with the configuration, wiring or sensor:

- 0x0001 Invalid Poles
- 0x0002 Amplitude Fail
- 0x0004 Resolver auto gain/phase active
- 0x0008 Bad magnitude

5.1.2 Control-Word (0x4632, sub 2)

In Genus, the controllers can support multi pole-pair resolvers and a range of resolver gains by manipulating the magnitude of the excitation waveform. The phase of the resolver must also be compensated for, in order to achieve optimum performance. Resolver gain and phase settings can now be auto-detected by Genus controllers by enabling the control words (auto-gain and auto-phase). This is described in more detail below.

- 0x0001 Auto-gain Test: Disabled by default (See section 5.2)
- 0x0002 Auto-phase test: Disabled by default (See section 5.3)

5.1.3 Resolver Pole Pairs (0x4632, sub 3)

Genus controllers can support multi pole-pair resolvers so enter the correct number of pole pairs here.

5.1.4 Excitation Magnitude (0x4632, sub 4)

This should be adjusted for different resolver gains. The results of the auto-gain test should be entered in this parameter (see section 5.2).

Example magnitudes: '0.29' gain resolver = 45%
'0.5' gain resolver = 25%

5.1.5 Excitation Phase Compensation (0x4632, sub 5)

For optimum performance, the phase of the resolver must also be compensated for so enter the result of the auto-tune phase result in this parameter (see section 5.3).

Typical phase compensation: 0 or 1

Some other debug parameters can be seen, including results from the auto-tuning tests.

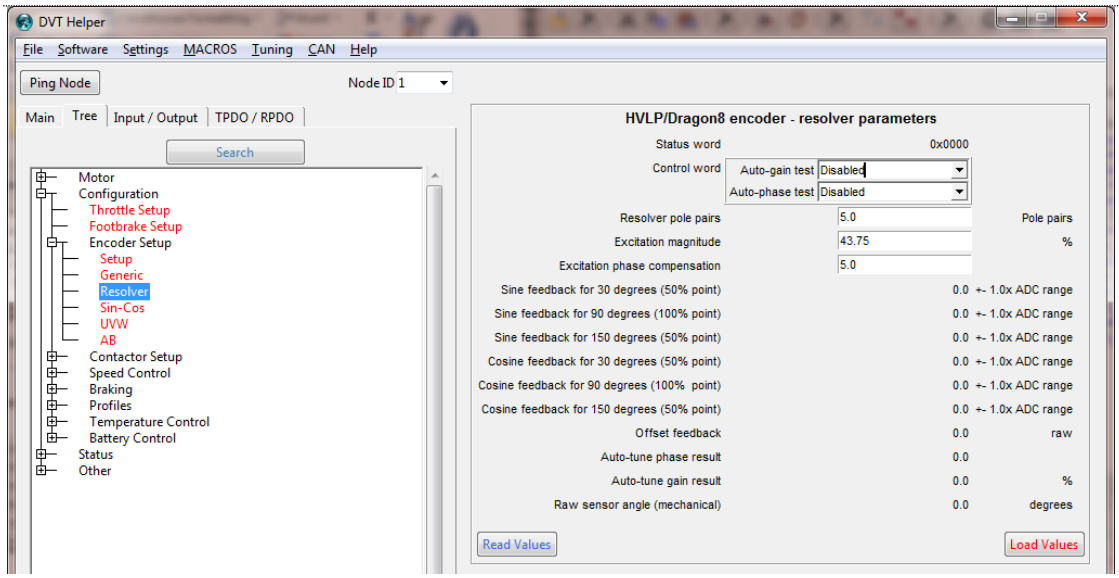


Figure 4 - Resolver settings in DVT Helper

5.2 Resolver gain and auto-gain test

The genus products allow manipulation of the magnitude of the resolver excitation signal to ensure that the returned signal from the resolver utilises the voltage range of the ADC converters.

A fault will be set if less than 25% of the ADC range is utilised (i.e. the magnitude is too small). A fault will also be set if 100% of the ADC range is utilised as this indicates signal saturation. Therefore the aim is to use around ±80-90% of the ADC range.

With the initial gain and phase parameters entered into 0x4632, to check the gain is appropriate, rotate the motor slowly. It is not necessary to control the motor to do this test – a hand rotation is sufficient.

- No faults should be set in the status word (in red).
  - 0x0001 Invalid Poles
  - 0x0002 Amplitude Fail
  - 0x0004 Resolver auto gain/phase active
- The feedback for the resolver sine 100% and cosine 100% values in 0x4632 should be measured at ±0.8

If neither are the case, then adjust the magnitude or perform an auto-gain test:

5.2.1 Auto-Gain test procedure

1. Ensure the motor is stationary. Motor control is NOT required for this test and will be prevented by an intentional fault being set to prevent drive.
2. Make sure you are logged in and in pre-operational mode.
3. Set the auto-gain bit of the resolver control-word (0x4632, 2) to be enabled by loading the value 0x0001 in DVT helper or changing it via SDO write.
4. Reading the control-word back should show the auto-gain test is enabled and status word (0x4632, 1) to 0x0004 as highlighted in figure 5. A fault is automatically set to ensure no motor control is possible in auto-gain mode.

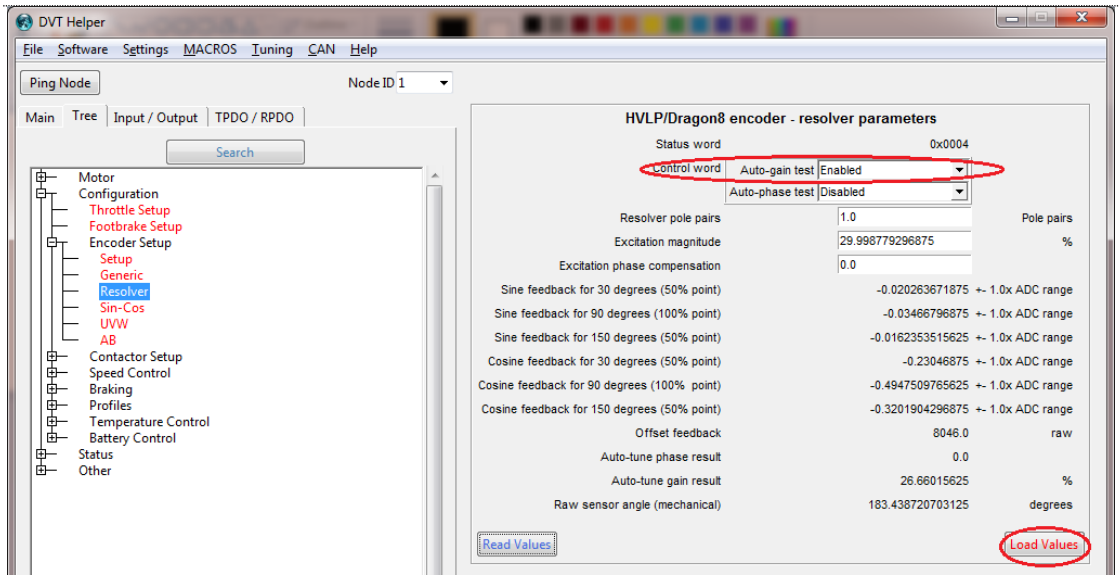


Figure 5 - Auto-gain enabled in resolver parameters in DVT helper

5. Observe the auto-tune gain result. Typically the following results are shown:
   '0.29' gain resolver = 45%
   '0.5' gain resolver = 25%.
6. Enter the suggested result into the Excitation magnitude item and disable the auto-gain test.
7. Using your hand, rotate the resolver manually. No resolver faults should be set (status word = 0x0000) and the feedback values for the sine and cosine 90% feedback points should move between ± 80% of the ADC range (± 10%)
8. Once the gain has been established, ensure the phase is correct or perform an auto-phase test (section 5.3).

In the example in the Figure 5, the auto-tune gain result has displayed 26.66015625. This suggests you can change the Excitation Magnitude (0x4632, 4) to 26 (Figure 6).

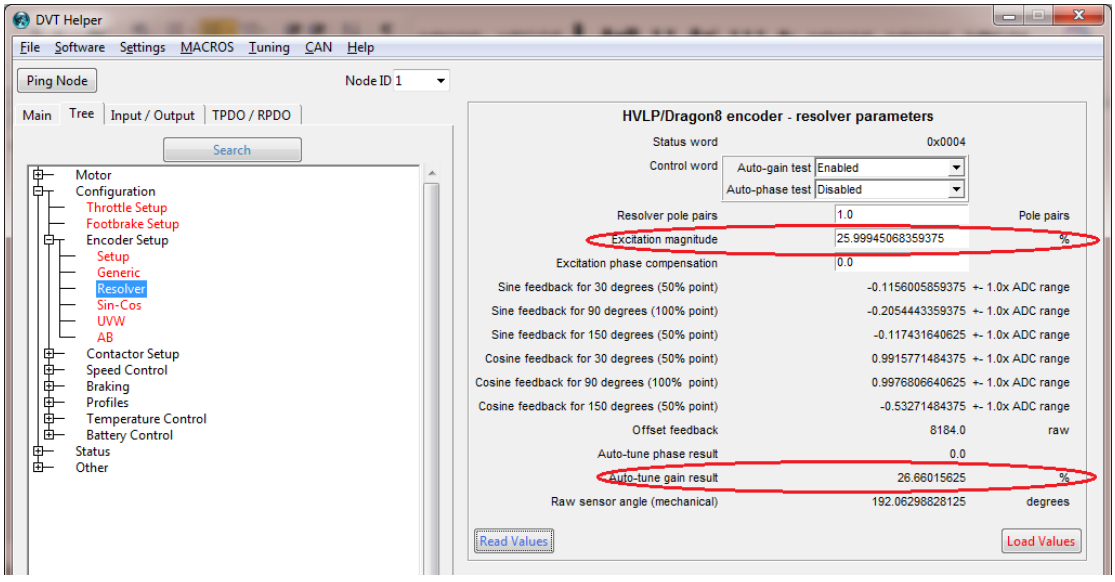


Figure 6 - Excitation magnitude and auto-tune gain result in DVT helper

5.3 Resolver phase and auto-phase test

For the resolver phase to be correct, the 50% sample points for the sine and cosine signals should be approximately half of the 100% sample points.

Figure 7 illustrates a resolver with an appropriate phase setting. For the particular angle of the resolver, the 50% points appear correct. Ensure auto-phase and gain test is disabled when checking the phase.

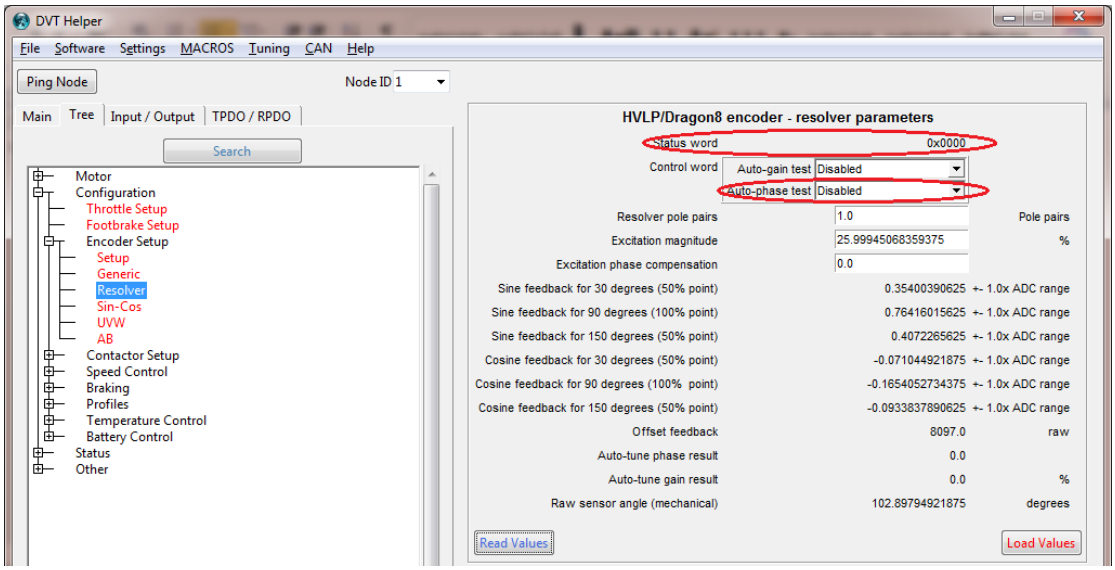


Figure 7 – Correct resolver gain

Rotate the motor to a few positions and check the 50% values appear correct. If not, then either adjust the phase or perform an auto-phase test.

5.3.1 Auto-Phase Test procedure

1. Ensure the motor is stationary. Motor control is NOT required for this test and will be prevented by an intentional fault bein set to prevent drive.
2. Make sure you are logged in and in pre-operational mode.
3. Set the Auto-phase test to enabled by setting the control-word (0x4632, 2) to 0x0002 or select “enabled” in the DVT helper.
4. Note: make sure the gain is correct and that an auto-gain test is not enabled.
5. The results from the auto-phase test can be seen in “auto-tune phase result” (0x4632, 13) as highlighted in Figure 8. Note, a value of zero is an acceptable answer.
6. The controller will fault during an auto-phase test to ensure no motor control is possible (status word = 0x0004).
7. Disable the auto-phase test.
8. Rotating the resolver by hand should and reading back the values should change the sine-cosine feedback and after a few seconds the auto-tune phase result may populate if it excitation phase compensation needs to be applied.
9. Once the test has completed, disable the test and load the values.
10. The status-word should return to 0x0000.
11. This completes the test.

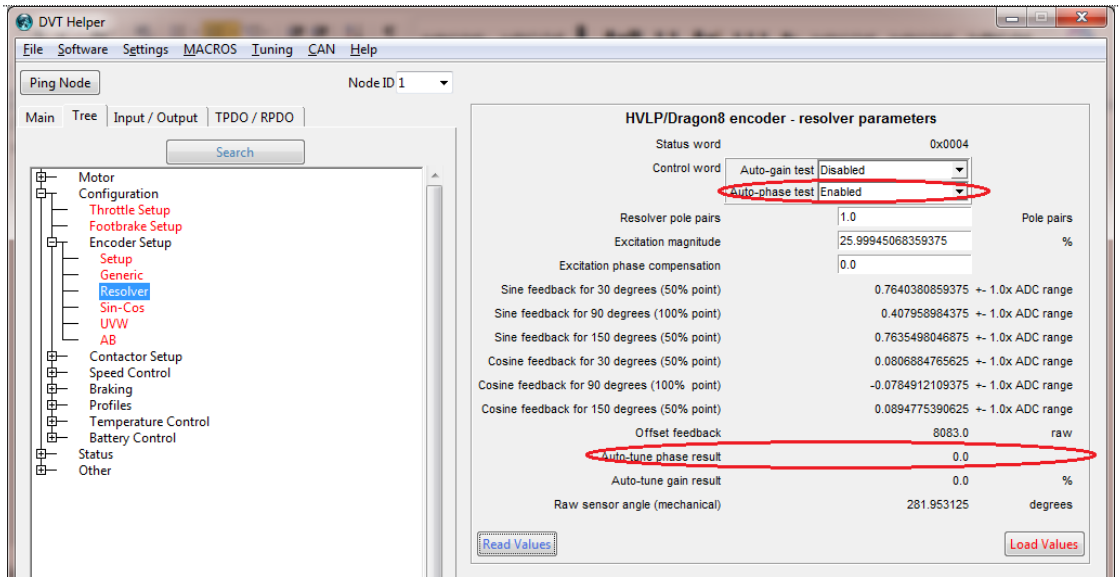


Figure 8 - Auto-tune phase test in the DVT helper

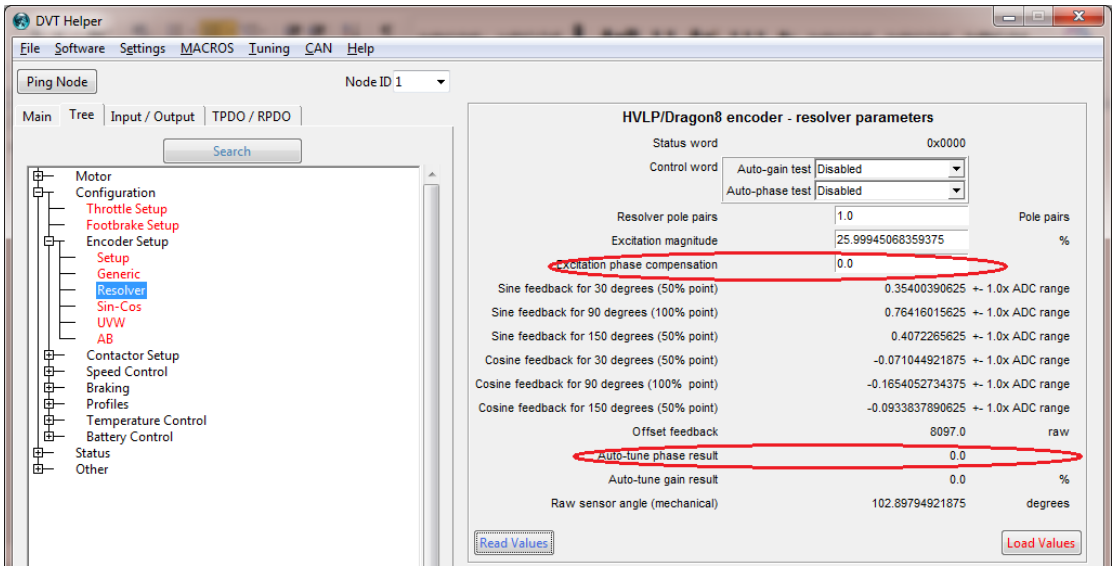


Figure 9 - Auto-tune phase result

6 Sine-Cosine (0x4633)

6.1 Parameters

The Sine-Cosine encoder settings reside in object 0x4632 and can also be modified in **Configuration -> Encoder Setup -> Sin-Cos** in DVT helper. Check that the encoder type is set as sine-cos in the encoder type in 0x4631, 1 (1 = Sine-Cosine).

6.1.1 Status-Word (0x4633, sub 1)

The status word is read only but will indicate either of the following states if there are issues with the configuration, wiring or sensor:

- 0x0001 Invalid waves per revolution
- 0x0002 Signal phase error
- 0x0004 Amplitude too small
- 0x0100 Warning for amplitude warning level (sub 12, 13) exceeded

6.1.2 Control-Word (0x4633, sub 2)

6.1.2.1 Sin/cos tracking disable (0x4633, sub 2, bit 0)

Min-max tracking can be disabled by writing 0x0001 (bit 0) to this object but it is enabled by default and recommended for most applications.

6.1.2.2 Sin/cos tracking values shown in sub 8-11 (0x4633, sub 2, bit 1)

To display the tracking of min/max in measured items set the object value to be 0x0002 (bit 1), this should only be done after the initial controller commissioning is done: cold values configured in sub 4-7 based on measured values in sub 8-11.

When setting 0x4633, sub2, bit1 to display tracking values in 0x4633, sub 8-11. These 4 objects can then be mapped to TPDOs to monitor the sin/cos sensor variation and the performance of the tracking algorithm.

6.1.3 Waves per Revolution (0x4633, sub 3)

The Genus controllers can support sine-cosine encoders with multiple waves-per-revolution and this can be entered here. Otherwise set the value here to 1.

6.1.4 Cold value sine/cosine minimum and maximum values (trough and peak) (0x4633, sub 4-7)

The minimum and maximum voltage levels of the sin/cos encoder must be entered in sub-indexes 4-7. During sensor commissioning the measured values in sub 8-11 should be copied to sub 4-7. Note 0x4633, sub 2 (sin/cos controlword) bit 1 should be cleared for this so that the tracking values are not being show in 0x4633, sub 8-11.

6.1.5 Measured value sine/cosine minimum and maximum values (trough and peak) (0x4633, sub 8-11)

To assist in configuration, the controller provides real-time feedback of the minimum and maximum sin/cos voltage levels in the measured values seen in sub 8 - 11. A rotating motor is required to determine these thresholds. These objects can display two different measurements:

- 0x4633, sub2, bit1 cleared = sub 8-11 display the raw measured sin/cos peak and trough values, specifically the lowest and highest values seen since the controller was last keyed on.
- 0x4633, sub2, bit1 set = sub 8-11 display the tracking values (assuming tracking hasn't been disabled using 0x4633, sub2, bit0) of sin/cos peak and trough values. Therefore the values can increase and decrease as the output of the sensor changes with temperature.

6.1.6 Warning level upper/lower threshold (0x4633, sub 12-13)

The voltages entered here cause a sin/cos sensor warning to be set if they are exceeded.

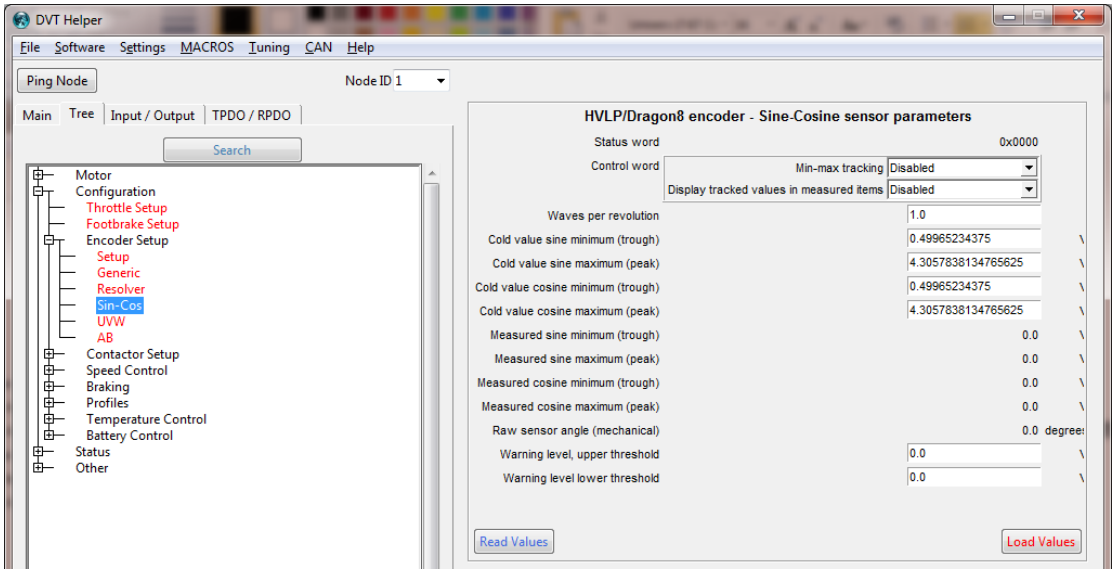


Figure 10 - Sine-Cosine sensor parameters

7 UVW (0x4634)

7.1 Parameters

The UVW configuration resides in object 0x4634. UVW encoders are commutation sensors designed to measure the absolute position of the rotor shaft within the motor, rather than its rotational speed. Rotor position information is used for control of permanent magnet motors, as it allows the controller to energise the motor phases appropriately based on the measured position of the magnets on the rotor.

Please note that UVW is a differential-input system on HVLP alpha requiring a differential signal pair for each UVW input. If it is desireable to use a single ended UVW sensor it is recommended to use HVLP beta instead.

For a PMAC motor, a full 360^0 sequence of UVW pulses must be present for each electrical cycle of the motor.

To configure the UVW encoder, set “UVW” as the encoder type in 0x4631, and then the UVW specific items are located in 0x4634:

7.1.1 Status-Word (0x4634, sub 1)

The status word is read only but will indicate either of the following states if there are issues with the configuration, wiring or sensor:

- 0x0001 Invalid state
- 0x0002 Invalid sequence
- 0x0004 Invalid motor pole pairs
- 0x0100 Status flag indicating that motor is stationary

7.1.2 Control-Word (0x4634, sub 2)

If the UVW inputs are inverted (active-low) then this inversion can be accounted for in the UVW invert flag 0x0001 (bit 0) in the control-word.

7.1.3 Pin State (0x4634, sub 3)

This is a debug item where bits 0-2 are the U, V, W pin states.

7.1.4 Last Sector time (0x4634, sub 4)

This is a debug item which is the last measured time between edges (DSP clocks).

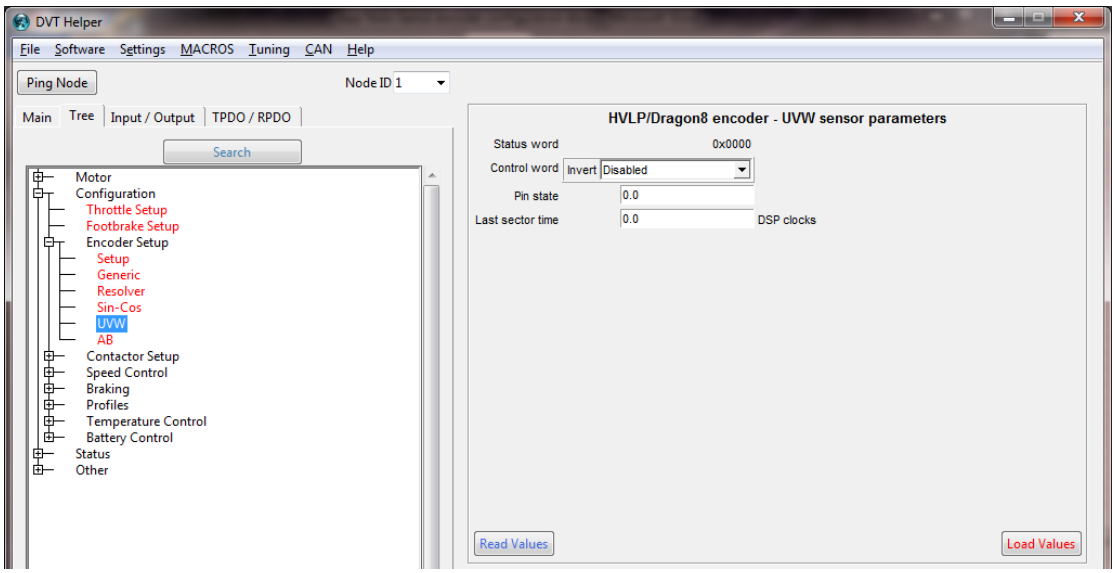


Figure 11 - UVW in DVT Helper

8 AB (0x4635)

8.1 Parameters

The AB encoder settings reside in object 0x4635 (Figure 12 AB in DVT Helper). A 6-wire connection is provided for open-collector quadrature pulse encoder devices. These types of encoder are optimized for accurate speed measurement, required for efficient control of induction motors.

Please note: that for HVLP alpha AB is a differential-input system requiring a 5V or 10V differential signal pair for each AB input. If it is desireable to use a single ended AB sensor it is recommended to use HVLP beta instead.

Please note: AB is not suitable for PMAC motors, where absolute position is required.

To configure the AB encoder, set "AB" as the encoder type in 0x4631, and then the AB specific items are located in 0x4635:

8.1.1 Status-Word (0x4635, sub 1)

The status word is read only but will indicate either of the following states if there are issues with the configuration:

- 0x0001 Invalid PPR (Pulses per revolution does not match, set this correctly in sub-index 3).
- 0x0002 Invalid motor pole pairs (Number of poles are wrong, set correctly in the generic encoder settings 0x4631)
- 0x0004 Wire-off (Check the encoder wiring)
- 0x1000 Status flag indicating that motor is stationary
- 0x2000 Invalid speed delta configuration (if supported)

8.1.2 Control-Word (0x4635, sub 2)

Setting 0x0001 (bit 0) enables AB tracking mode (only supported if object for result 0x4635, sub 5 is present). This can be used to compensate for phase offsets between A and B. The result of the phase offset measurement is displayed in 0x4635, sub 5

8.1.3 Pulses per Revolution (0x4635, sub 3)

The pulses per revolution must be set to match the encoder. If this is incorrect, the status word in sub 1 will change to 0x0001.

8.1.4 AB Counter (0x4635, sub 4)

This is a debug item that measures the time between DSP edges.

8.1.5 Enable speed delta check (0x4635, sub 6)

Boolean flag to enable wire-off speed delta check

8.1.6 AB wire-off speed delta (0x4635, sub 7)

Threshold for setting wire-off fault from speed delta. Specified in krpm/sec

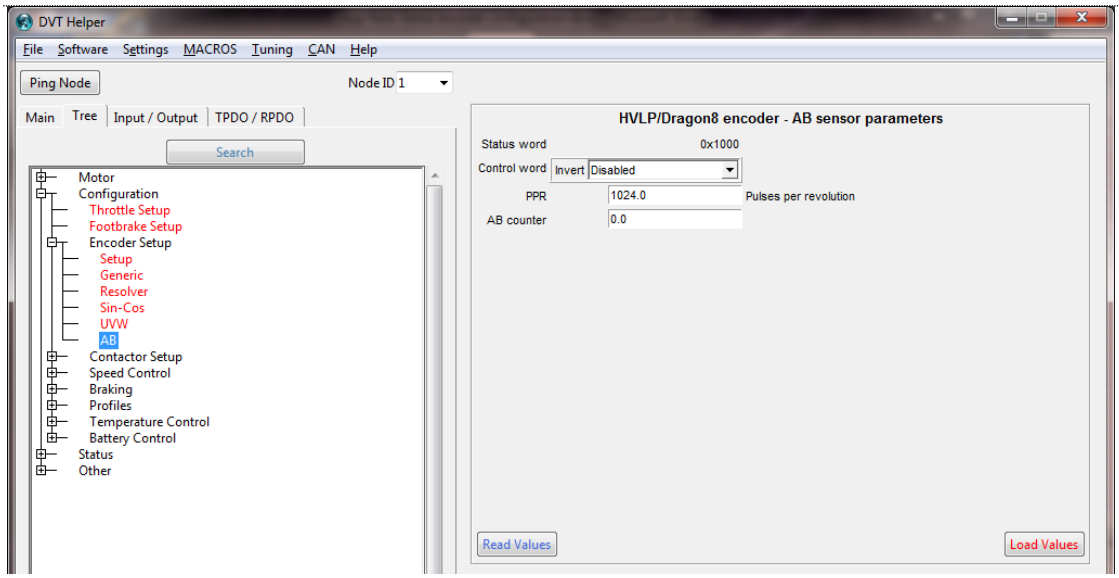


Figure 12 - AB in DVT Helper

9 Encoder Auto-Alignment

9.1 Background

Genus contains a new facility to measure the alignment of an absolute position encoder for PMAC motors. This is a low-speed automated test in which the motor is slowly rotated and the encoder offset measured. However, this test is not fully automatic. The test must be enabled, apply torque and allow the test to complete. After this is completed, the results should be copied into the encoder offset parameter. The intention of the test is to provide an approximate angle, sufficient to allow rotation. Fine-tuning should still be performed by conventional alignment methods.

Latency effects (e.g. for Sin-Cos or UVW at high speeds) must also be tuned-out by the application engineer, but the low speed nature of this alignment test won't be impacted by any latency.

9.2 Before running the test

- Ensure you are logged in and in pre-operational.
- The motor must be relatively free to rotate because while the user-specified torque will allow the motor to overcome some frictional load, considerable load friction will affect the quality of the result or cause the test to abort.
- The encoder offset in 0x4631 should be zeroed before doing the alignment test; otherwise the result will be an offset from the existing offset.
- Ensure encoder latency is at a known value or zero.
- Ensure encoder PLL filter is at known values or OBD default values.
- Ensure basic PMAC motor parameters are correct, such as
    - Motor pole-pairs
    - Encoder pole pairs
    - Ke is semi-sensible (not 10x too big)
    - PI gains are sensible (so control faults do not occur)
- It is important to note that the auto-alignment is an absolute encoder alignment intended for geometric PMAC. Set the motor type (0x4639, 0 to 0x0011). The alignment will also work with lookup-table PMAC control (0x4639, 0 to 0x0012); although this requires the lookup tables are sensible for the motor. Typically at this stage of system configuration, the motor tables will not be known, so standard geometric PMAC mode is recommended.

The auto-alignment is not encoder-type specific and will attempt to operate on any selected encoder – i.e. Resolver, Sin-Cos, UVW or AB so follow the specific instructions for the correct encoder type below:

- If using a Resolver ensure pole-pairs, gain and phase are set correctly before performing the auto-alignment. Gain and phase can be auto-tested with a stationary unpowered motor.
- If using Sin-Cos, ensure waves-per-revolution and min/max voltages are set correctly before performing the auto-alignment. These can be obtained by rotating the motor by external means or via datasheet.
- UVW cannot determine a low-speed angle to a precision better than 60 degrees; the accuracy of the auto-alignment will be limited by this, so the auto-alignment procedure is not recommended.
- AB is not a suitable encoder for PMAC motors as it is not an absolute angle sensor, so any results from an auto-alignment should be ignored and an appropriate sensor used for PMAC or for induction motors the offset set to 0.

9.3 Important to Note

When enabled, the test will repeat indefinitely. Results will only be generated if sufficient movement is detected, otherwise the test will abort and restart until motion can be achieved. If the motor rotates, but the test repeatedly aborts (see procedure), ensure the applied torque is sufficient for rotation but does not cause excessive oscillations. Also, check the parameters for the selected encoder (e.g. pole pairs). Ensure that currents are following their references as current control faults may not trigger for low torque demands. An inability to follow current references suggests either completely insufficient current control gains or absent motor phase connections.

9.4 Procedure

1. Check the Resolver phase and magnitude are sensible by running the auto-gain tests first and also check Sin-cos min and max values are sensible.
2. Selected "Enabled" in **Encoder Setup -> Generic -> Primary Encoder Control Word -> Encoder auto-align test** as highlighted below. This is equivalent to setting bit 3 (0x0008) of object 0x4631, 11. After loading these values perform a key-recycle.

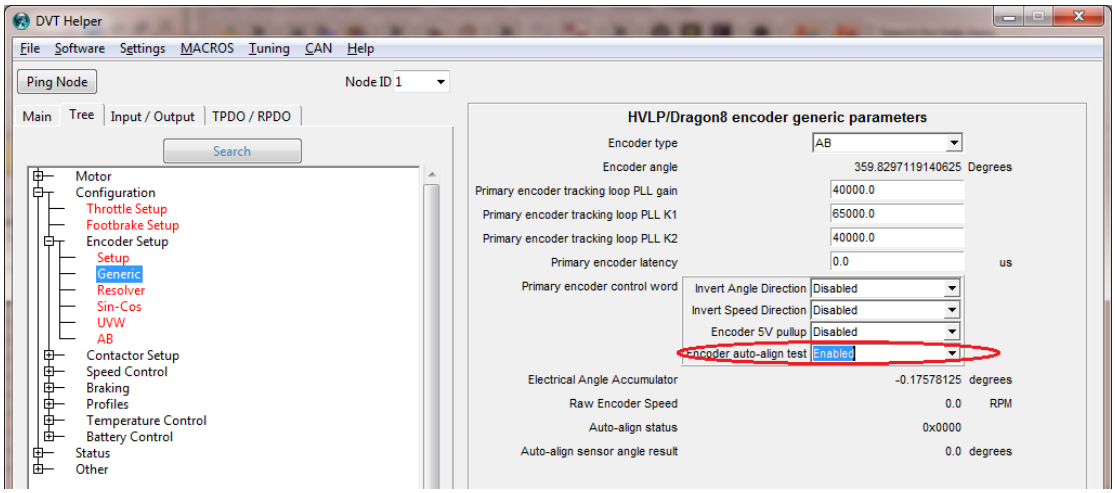


Figure 13 – enabling encoder align mode

3. You should now be able to see the Encoder Angle (0x4631, 3) moving, even for a stationary motor – this is OK and a feature of the test.
4. Zero the existing encoder offset in **Encoder Setup -> Setup -> Encoder offset** (object 0x4630, 4) as highlighted below.

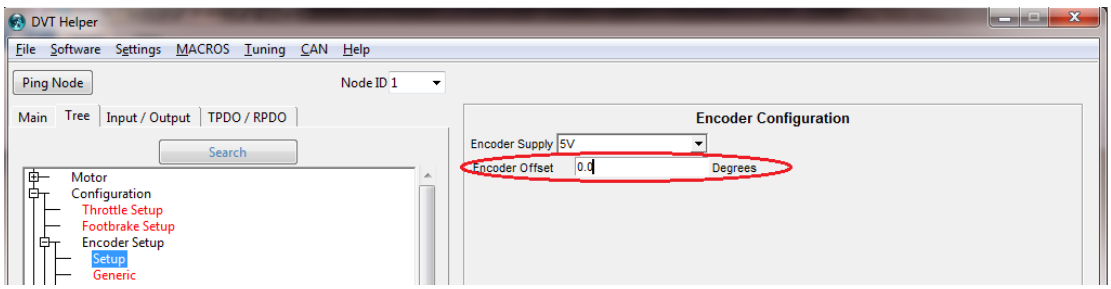


Figure 14 – setting the new encoder offset

5. Ensure the DC link is present and then enable the controller.
6. Demand a suitable torque – typically 5% of maximum stator current which can be found in the **Motor -> Motor Data -> Maximum stator current** tree e.g. 45 amps is 2.25.
7. Observe the motor – it should be slowly rotating (it will move clockwise and anticlockwise for a test cycle).

8. Observe Auto-align status (0x4631, 15). Bits 0-3 will show a counter. This will count from 0x00 to 0x0f for a complete test. The counter will restart around 0x04 if the test cannot complete (i.e. insufficient torque).
9. Observe Auto-align status (0x4631, 15). If bit 9 (0x0200) is set, then flip the angle direction, either by crossing two motor connection phases, or via software (set bit 0 in 0x4641,11) and repeat the test. This bit should now remain clear.
10. Observe Auto-align status (0x4631, 15). When bit 8 (0x0100) toggles, a new result is present.
11. Observe the result in Auto-align sensor angle result (0x4631, 16). If the result remains consistent after a few passes, copy the value into the Encoder Offset (**Encoder Setup -> Setup -> Encoder offset**) 0x4630, 4).
12. Verify the quality of the offset by analysing voltage vectors during a run-down test (the standard procedure).

10 Encoder Fault Codes (EMCY message)

When an encoder fault occurs, it is possible to obtain extra information about the fault by looking at the Fault Data Bytes in the EMCY message.

The first byte should be 0xFF, and the second byte should be 0x03. The third byte represents the value of internal variable angle_fault. The value of this depends on the type of encoder that is configured.

10.1.1 Resolver

- 0x01 Invalid poles configuration
  - Motor and resolver pole pairs must be greater than 0 and less than 128
  - Resolver pole pairs must be multiple of motor pole pairs
- 0x02 Amplitude fail
  - Resolver sin/cos values at waveform peaks are used to calculate $\sin^2 + \cos^2$. Must be in the per-unit range 0.06 to 1.2
- 0x04 Resolver auto-tune mode active
  - Fault set to prevent operation during resolver auto-tune
- 0x08 Bad magnitude
  - All resolver sin/cos samples are used to calculate $\sin^2 + \cos^2$. This must be larger than 37%

10.1.2 Sin-Cos

- 0x01 Invalid motor poles / waves per revolution configuration
  - Motor pole pairs and sin-cos waves per revolution must be greater than 0 and less than 255
  - Sin-cos waves per revolution must be multiple of motor pole pairs
- 0x02 Signal phase error
  - The per-unit values of sin and cos are used to calculate $\sin^2 + \cos^2$. Values outside of the range 0.7 to 1.5
- 0x04 Amplitude too small
  - Peak and trough of measured sin-cos signals must be larger than 1volt

10.1.3 UVW

- 0x01 Invalid state of UVW signals
  - All of UVW high or all UVW low
- 0x02 Invalid sequence
  - Transition of signals should follow the pattern U -> V -> W (or reverse for negative direction)
- 0x04 Invalid motor pole pair configuration
  - Motor pole pairs must be non-zero

10.1.4 AB

- 0x01 Invalid ppr configuration
  - AB ppr must be at least 16
- 0x02 Invalid motor pole-pair configuration
  - Motor pole pairs must be non-zero
- 0x04 Wire-off detected
  - AB edge count compared to time between edges suggest wire-off